Discrete vs Continuous Variables

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Introduction

Random variables come in two canonical kinds – **discrete** (countable support) and **continuous** (uncountable support with a density). Most textbook distributions belong to one or the other. A subtler third category, **mixed**, combines discrete atoms with continuous parts and arises naturally in censored, truncated, or zero-inflated data.

Prerequisites

Random variables, PMF, and PDF.

Theory

A random variable X is:

- **Discrete** if it takes values in a countable set $\{x_1, x_2, \ldots\}$. Its distribution is described by the **probability mass function** $p(x_i) = P(X = x_i)$, with $\sum_i p(x_i) = 1$.
- **Continuous** (absolutely continuous) if there exists a **probability density function** f such that $P(X \in A) = \int_A f(x)\,dx$ for every measurable set A. For continuous X, $P(X = x) = 0$ for every single point x.
- **Mixed** if its distribution has both discrete atoms (points with positive probability mass) and a continuous part. Examples: censored survival times (mass at the censoring time, density elsewhere); zero-inflated counts (mass at 0, positive distribution above).

The CDF $F_X(x) = P(X \le x)$ is defined for all three. For discrete X, F is a step function; for continuous X, F is smooth; for mixed X, F is piecewise smooth with jumps at the atoms.

The mean and variance are defined the same way (as $E[X]$ and $E[(X - E[X])^2]$) across the three cases, computed via sums for discrete and integrals for continuous.

Assumptions

The classification is mathematical rather than empirical; *any* random variable falls into one of the three categories.

R Implementation

```r
library(ggplot2)
set.seed(2026)

# Discrete: Poisson counts
dis <- rpois(5000, lambda = 3)
table(dis)[1:7]

# Continuous: exponential waiting times
con <- rexp(5000, rate = 0.5)
quantile(con, c(0.25, 0.5, 0.75))
```

```r
# Mixed: censored normal (values below 0 set to exactly 0)
mix <- pmax(rnorm(5000, mean = 1.0, sd = 1.5), 0)
c(mass_at_zero = mean(mix == 0),
  mean_overall = mean(mix))

# Plot all three
df <- rbind(
  data.frame(x = dis,    kind = "discrete (Poisson)"),
  data.frame(x = con,    kind = "continuous (exponential)"),
  data.frame(x = mix,    kind = "mixed (censored normal)")
)

ggplot(df, aes(x = x)) +
  geom_histogram(bins = 40, fill = "#2A9D8F") +
  facet_wrap(~ kind, scales = "free", ncol = 1) +
  theme_minimal()
```

Output & Results

```
dis       0      1      2      3      4      5      6
count   243    746   1124   1128    835    505    275

      25%       50%       75%
 0.5675    1.3735    2.7711

mass_at_zero  mean_overall
      0.2526        1.1342
```

The Poisson histogram has bars at integer values (discrete), the exponential has a smooth decaying curve (continuous), and the censored-normal has a spike at zero plus a continuous upper tail (mixed).

Interpretation

In reporting, the kind of random variable determines appropriate summaries:

- Discrete: report counts, proportions, mode, or a PMF table.
- Continuous: report mean, SD, quantiles, or a histogram/density.
- Mixed: report the atom probability separately from the continuous-part summary.

A paper analysing a zero-inflated outcome should acknowledge the mixed distribution explicitly rather than forcing it into a continuous or discrete mould.

Practical Tips

- Count data that "cap out" at a large value are often mixed; report the probability of the cap and analyse the remainder conditionally.
- Censored survival times are mixed; classical survival methods (Kaplan-Meier, Cox) handle the mixture natively.
- Continuous variables that are rounded to, say, integer values are still "almost continuous" in practice; treat as continuous for inference.
- Zero-inflated Poisson (`pscl::zeroinfl`) is the canonical regression model for mixed count data.
- When in doubt, plot a histogram (or ECDF) and inspect for atoms; the presence of a spike at a specific value signals a mixed distribution.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots